Solutions to Exam HT23
SF2940 Probability Theory

Problem 1

(a)

By definition, $X_n \xrightarrow{L^2} X$ as $n \to \infty$ if

$$\mathbb{E}\left[(X_n - X)^2\right] \xrightarrow[n\to\infty]{} 0.$$

(b)

For any $\epsilon > 0$, we have by Markov's / Chebyshev's inequality that

$$P(|X_n - X| > \epsilon) \leq \frac{\mathbb{E}\left[(X_n - X)^2\right]}{\epsilon^2}.$$

Since $X_n \xrightarrow{L^2} X$ as $n \to \infty$, we conclude that

$$\lim_{n\to\infty} P(|X_n - X| > \epsilon) = 0.$$

(c)

If $X \sim \text{Exp}(1)$, then $f_X(x) = e^{-x}\mathbf{1}_{\{x>0\}}$. The moment generating function $\psi_X(t)$ is given by

$$\psi_X(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx}e^{-x}\,dx = \int_0^\infty e^{-(1-t)x}\,dx = \frac{1}{1-t},$$

for all $|t| < 1$.

(d)

The k-th moment of X is given by

$$\mathbb{E}[X^k] = \psi_X^{(k)}(0) = \left.\frac{d^k}{dt^k}\psi_X(t)\right|_{t=0}.$$

From part (c), for $|t| < 1$:

$$\begin{aligned}
\psi_X'(t) &= \frac{1}{(1-t)^2}, \\
\psi_X''(t) &= \frac{2}{(1-t)^3}, \\
\psi_X'''(t) &= \frac{2\cdot 3}{(1-t)^4}, \\
&\vdots \\
\psi_X^{(k)}(t) &= \frac{1\cdot 2\cdot 3\cdots k}{(1-t)^{k+1}} = \frac{k!}{(1-t)^{k+1}}.
\end{aligned}$$

Setting $t = 0$ above, we find $\mathbb{E}[X^k] = k!$.

Alternatively: For $|t| < 1$, expanding the exponential function gives

$$\psi_X(t) = \mathbb{E}[e^{tX}] = 1 + \sum_{k=1}^{\infty} \frac{t^k}{k!} \mathbb{E}[X^k].$$

On the other hand, via the geometric series:

$$\psi_X(t) = \frac{1}{1-t} = 1 + \sum_{k=1}^{\infty} t^k.$$

Comparing coefficients of t^k, we must have $\frac{\mathbb{E}[X^k]}{k!} = 1 \implies \mathbb{E}[X^k] = k!$.

Problem 2

(a)

We have by definition:

$$f_{X|Y=y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

The marginal density $f_Y(y)$ is given, for $y > 0$, by

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx = 4\int_0^y e^{-2y}\,dx = 4ye^{-2y}.$$

Thus,

$$f_{X|Y=y}(x) = \frac{4e^{-2y}\mathbf{1}_{\{0<x<y\}}}{4ye^{-2y}} = \frac{1}{y}\mathbf{1}_{\{0<x<y\}}.$$

Therefore, $X \mid Y = y$ is uniformly distributed on the interval $(0, y)$, and we have

$$\mathbb{E}[X \mid Y] = \frac{Y}{2}.$$

(b)

We set

$$U := Y, \quad V := \frac{X}{Y - X}.$$

The density $f_{X,Y}(x,y)$ is supported on $S := \{(x,y) \in \mathbb{R}^2 : 0 < x < y\}$. Let

$$g_1(x,y) := y, \quad g_2(x,y) := \frac{x}{y-x}.$$

Then $\mathbf{g}\colon S \to T$ maps $(x,y) \mapsto (u,v)$ with $T := \{(u,v) \in \mathbb{R}^2 : u > 0, v > 0\}$. The inverse map $\mathbf{h} = (h_1, h_2)\colon T \to S$ is given by

$$(u,v) \mapsto \left(\frac{uv}{v+1}, u\right), \quad \text{i.e., } h_1(u,v) = \frac{uv}{v+1}, \quad h_2(u,v) = u.$$

The Jacobian of the transformation is

$$J = \det\begin{pmatrix} \frac{\partial h_1}{\partial u} & \frac{\partial h_1}{\partial v} \\ \frac{\partial h_2}{\partial u} & \frac{\partial h_2}{\partial v} \end{pmatrix} = \det\begin{pmatrix} \frac{v}{v+1} & \frac{u}{(v+1)^2} \\ 1 & 0 \end{pmatrix} = -\frac{u}{(v+1)^2}.$$

By the transformation theorem:

$$\begin{aligned} f_{U,V}(u,v) &= f_{X,Y}\left(\frac{uv}{v+1}, u\right)|J| \\ &= 4e^{-2u}\frac{u}{(1+v)^2}\mathbf{1}_{\left\{0<\frac{uv}{v+1}<u\right\}}\mathbf{1}_{\{u>0,v>0\}}. \end{aligned}$$

Since for $u > 0$ and $v > 0$, the inequality $0 < \frac{v}{v+1} < 1$ always holds, the condition $0 < \frac{uv}{v+1} < u$ is automatically satisfied. Thus,

$$f_{U,V}(u,v) = 4ue^{-2u}\mathbf{1}_{\{u>0\}} \cdot \frac{1}{(1+v)^2}\mathbf{1}_{\{v>0\}}.$$

Remark: The joint density $f_{U,V}(u,v)$ factorizes into the product of the marginals $f_U(u)$ and $f_V(v)$; hence U and V are independent. In fact, $U \sim \Gamma\left(2, \frac{1}{2}\right)$ and $V \sim F(2,2)$.

Problem 3

(a)

We have $\mathbf{Y} = B\mathbf{X}$ with

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix}.$$

By the linear transformation theorem, $\Lambda_{\mathbf{Y}} = B\Lambda_{\mathbf{X}}B^T$:

$$\Lambda_{\mathbf{Y}} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 3 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 9 & 2 \\ 2 & 2 \end{pmatrix}.$$

Hence $\mathbf{Y} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 9 & 2 \\ 2 & 2 \end{pmatrix}\right)$, since $\mathbb{E}[\mathbf{Y}] = B\mathbb{E}[\mathbf{X}] = \mathbf{0}$.

(b)

The conditional density is

$$f_{Y_1|Y_2=0}(y_1) = \frac{f_{Y_1,Y_2}(y_1, 0)}{f_{Y_2}(0)}.$$

Since $Y_2 \sim \mathcal{N}(0, 2)$, we have

$$f_{Y_2}(0) = \frac{1}{\sqrt{2\pi \cdot 2}} = \frac{1}{\sqrt{4\pi}}.$$

To determine $f_{Y_1,Y_2}(y_1, y_2)$, we invert $\Lambda_{\mathbf{Y}}$:

$$\det(\Lambda_{\mathbf{Y}}) = 9 \cdot 2 - 2^2 = 14 \implies \Lambda_{\mathbf{Y}}^{-1} = \frac{1}{14}\begin{pmatrix} 2 & -2 \\ -2 & 9 \end{pmatrix}.$$

Thus,

$$f_{Y_1,Y_2}(y_1, y_2) = \frac{1}{2\pi\sqrt{14}} \exp\left(-\frac{1}{2}(y_1, y_2)\Lambda_{\mathbf{Y}}^{-1}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right).$$

Setting $y_2 = 0$:

$$f_{Y_1,Y_2}(y_1, 0) = \frac{1}{2\pi\sqrt{14}} \exp\left(-\frac{1}{2} \cdot \frac{2}{14} y_1^2\right) = \frac{1}{2\pi\sqrt{14}} e^{-\frac{1}{14}y_1^2}.$$

Therefore,

$$f_{Y_1|Y_2=0}(y_1) = \frac{\frac{1}{2\pi\sqrt{14}} e^{-\frac{1}{14}y_1^2}}{\frac{1}{\sqrt{4\pi}}} = \frac{1}{\sqrt{2\pi \cdot \frac{7}{2}}} e^{-\frac{y_1^2}{2\cdot(7/2)}} = \frac{1}{\sqrt{7\pi}} e^{-\frac{1}{14}y_1^2}.$$

(That is, $Y_1 \mid Y_2 = 0 \sim \mathcal{N}\left(0, \frac{7}{2}\right)$).

(c)

Using the identity $\max\{a, b\} = \frac{a+b}{2} + \frac{|a-b|}{2}$:

$$\max\{X_1, X_3\} = \frac{X_1 + X_3}{2} + \frac{|X_1 - X_3|}{2}.$$

Taking expectations (noting that $\mathbb{E}[X_1 + X_3] = 0$):

$$\mathbb{E}[\max\{X_1, X_3\}] = \frac{\mathbb{E}[|X_1 - X_3|]}{2} = \frac{\mathbb{E}[|Y_2|]}{2}.$$

Since $Y_2 = X_1 - X_3 \sim \mathcal{N}(0, 2)$:

$$\begin{aligned}\mathbb{E}[|Y_2|] &= \frac{1}{\sqrt{4\pi}} \int_{-\infty}^{\infty} |y_2| e^{-\frac{1}{4}y_2^2}\, dy_2 = \frac{2}{\sqrt{4\pi}} \int_0^{\infty} y_2 e^{-\frac{1}{4}y_2^2}\, dy_2 \\ &= \frac{-2}{\sqrt{4\pi}} \int_0^{\infty} 2\frac{d}{dy_2}\left(e^{-\frac{1}{4}y_2^2}\right) dy_2 = -\frac{4}{\sqrt{4\pi}}\left[e^{-\frac{1}{4}y_2^2}\right]_0^{\infty} \\ &= -\frac{4}{\sqrt{4\pi}}(0-1) = \frac{4}{2\sqrt{\pi}} = \frac{2}{\sqrt{\pi}}.\end{aligned}$$

Hence,

$$\mathbb{E}[\max\{X_1, X_3\}] = \frac{1}{2} \cdot \frac{2}{\sqrt{\pi}} = \frac{1}{\sqrt{\pi}}.$$

Problem 4

(a)

First, we note that

$$\{M_n > y\} = \{X_1 > y, X_2 > y, \ldots, X_n > y\}.$$

By independence and identical distribution of $X_1, \ldots, X_n$:

$$P(M_n > y) = \prod_{i=1}^{n} P(X_i > y) = (P(X_1 > y))^n.$$

For $y \in (0, 1)$:

$$P(X_1 > y) = 1 - P(X_1 \leq y) = 1 - 2\int_0^y (1 - x)\, dx = 1 - 2\left(y - \frac{y^2}{2}\right) = 1 - 2y + y^2 = (1 - y)^2.$$

(Note that $P(X_1 > y) = 0$ if $y \geq 1$ and $P(X_1 > y) = 1$ if $y \leq 0$).

Since $M_n \geq 0$, for any $\epsilon \in (0, 1)$ we have:

$$P(|M_n - 0| > \epsilon) = P(M_n > \epsilon) = (P(X_1 > \epsilon))^n = (1 - \epsilon)^{2n}.$$

(For $\epsilon \geq 1$, $P(|M_n| > \epsilon) = 0$). Since $0 < 1 - \epsilon < 1$, $(1 - \epsilon)^{2n} \to 0$ as $n \to \infty$. Thus,

$$M_n \xrightarrow[n\to\infty]{\mathbb{P}} 0.$$

(b)

For fixed $y > 0$ and sufficiently large n:

$$\begin{aligned} P(nM_n \leq y) &= P\left(M_n \leq \frac{y}{n}\right) = 1 - P\left(M_n > \frac{y}{n}\right) \\ &= 1 - \left(P\left(X_1 > \frac{y}{n}\right)\right)^n = 1 - \left(1 - \frac{y}{n}\right)^{2n}. \end{aligned}$$

By Euler's formula for the exponential function, $\lim_{n\to\infty} \left(1 - \frac{y}{n}\right)^{2n} = e^{-2y}$. Thus,

$$F_{nM_n}(y) \xrightarrow[n\to\infty]{} F_Y(y) = \begin{cases} 1 - e^{-2y}, & y > 0, \\ 0, & y \leq 0, \end{cases}$$

which is the distribution function of $Y \sim \text{Exp}(2)$ (or $\text{Exp}(\frac{1}{2})$ with mean $\frac{1}{2}$). Since the limiting CDF is continuous everywhere, $nM_n \xrightarrow{d} Y$.

(c)

Let $H_n := \{N = n\}$ for $n = 1, 2, \ldots$. Then $\bigcup_{n=1}^{\infty} H_n = \Omega$ and $H_n \cap H_m = \emptyset$ for $n \neq m$.

Let $A := \{M_N > y\}$. By the law of total probability (σ-additivity):

$$\begin{aligned} P(A) &= P\left(\bigcup_{n=1}^{\infty} (A \cap H_n)\right) = \sum_{n=1}^{\infty} P(A \cap H_n) \\ &= \sum_{n=1}^{\infty} P(A \mid H_n) P(H_n) = \sum_{n=1}^{\infty} P(M_n > y \mid N = n) P(N = n). \end{aligned}$$

Since N is independent of $(X_i)_{i\geq 1}$, this simplifies to:

$$P(M_N > y) = \sum_{n=1}^{\infty} P(M_n > y)P(N = n).$$

By part (a), $P(M_n > y) = (P(X_1 > y))^n$. Thus,

$$P(M_N > y) = \sum_{n=1}^{\infty} (P(X_1 > y))^n P(N = n) = g_N(P(X_1 > y)),$$

where $g_N(t) = \sum_{n=0}^{\infty} t^n P(N = n)$ is the probability generating function of N (noting that $P(N = 0) = 0$). Consequently,

$$P(M_N \leq y) = 1 - g_N(P(X_1 > y)).$$